AC Type-2 Charging - I

Introduction

The rapid adoption of electric vehicles (EVs) necessitates robust and reliable charging infrastructure. While DC fast charging often grabs headlines for its speed, AC charging remains the backbone for everyday use, particularly for overnight charging at homes and workplaces. This lecture delves into the intricacies of AC Type-2 charging systems, a prevalent standard, and explores how vehicles and charging stations communicate to ensure safe and efficient power delivery. Understanding the underlying architecture, communication protocols, and safety mechanisms is crucial for anyone involved in the design, deployment, or maintenance of EV charging solutions.

This session specifically focuses on the AC charging process, highlighting the roles of the Electric Vehicle Supply Equipment (EVSE) and the vehicle's onboard charger. We will examine the communication pathways that facilitate a secure connection, from the initial plug-in detection to the negotiation of charging parameters. By dissecting the components and their interactions, we gain insight into how AC charging, despite being a "slower" method, is a sophisticated interplay of electrical engineering and communication protocols designed for user safety and battery longevity.

Core Concepts

Term	Definition	Significance
`ACVSE`	Electric Vehicle Supply Equipment, the external charging station unit that connects to the grid.	Provides the AC power interface and manages communication with the EV and central systems.
`Onboard Charger`	A power conversion unit located within the EV, comprising AC-DC and isolated DC-DC converters.	Converts incoming AC power from the grid into DC power suitable for the battery, ensuring isolation and proper voltage/current.
`BMS`	Battery Management System, an electronic system that	Acts as the "master" during charging, communicating

Term	Definition	Significance
	monitors and manages the EV's battery pack.	battery status and charging readiness to the ACVSE.
`CP Pin`	Control Pilot pin, a dedicated pin in the charging connector used for PWM-based communication.	Facilitates low-level communication between the EV and ACVSE to manage the charging sequence.
`PP Pin`	Proximity Pilot pin, a dedicated pin in the charging connector used for detecting plug-in and cable current capacity.	Ensures a correct and safe connection by identifying the charger plug and its maximum current rating.
`PWM Communication`	Pulse Width Modulation-based communication, a method where information is encoded by varying the duty cycle or amplitude of a square wave.	Enables low-level, robust communication between the EV and ACVSE, especially for AC charging.
`R6 Resistance`	A specific resistance value located within the charger plug's PP circuit.	Crucial for the EV to determine the maximum current capacity of the connected charging cable and plug type.
`CCS2`	Combined Charging System Type 2, a universal connector standard that supports both AC and DC charging.	Allows for versatile charging options using a single vehicle inlet, with the top portion compatible with AC Type-2 plugs.

Detailed Analysis

1. AC Charging System Architecture and Power Flow

The AC charging system is fundamentally divided into components residing within the Electric Vehicle Supply Equipment (EVSE) and those integrated into the Electric

Vehicle (EV). Unlike DC charging where significant power conversion happens outside the vehicle, AC charging relies heavily on the EV's internal systems.

- **EVSE (ACVSE) Components**: The ACVSE, often a wall-mounted unit, primarily consists of a controller, sensors, and safety interlocks.
- **Controller**: Manages communication with the EV's Battery Management System (BMS) and, in commercial settings, with a central management system for metering and billing.
- **Sensors**: Monitor voltage and current for protection and monitoring purposes.
- **Safety Interlocks**: Comprising relays and contactors, these ensure immediate disconnection from the grid in case of a fault on either the EV or EVSE side, prioritizing user safety.
- **Power Delivery**: The ACVSE directly supplies AC power (single-phase or three-phase) from the grid to the vehicle via the charging cable and plug.
- **EV Internal Components**: The vehicle houses the critical power conversion and management units.
- **Onboard Charger**: This is the heart of AC charging within the EV. It comprises an AC-DC converter followed by an isolated DC-DC converter.
- The AC-DC converter rectifies the incoming AC power.
- The isolated DC-DC converter provides galvanic isolation between the battery pack and the AC grid, and regulates the DC voltage and current to the precise levels required for battery charging.
- Since the onboard charger controls the final voltage and current, it dictates the charging profile.
- **Battery Management System (BMS)**: The BMS continuously monitors the battery pack's voltage, current, and temperature. It determines the battery's State of Charge (SoC) and State of Health (SoH).
- The BMS acts as the "master" in the communication process, sending critical information to the ACVSE regarding the vehicle's readiness and charging requirements.
- This master role means the EV, specifically its BMS, decides when to start and stop charging, ensuring optimal battery health and safety.

2. Communication Protocols in AC Charging

Communication is paramount for safe and efficient AC charging. Given that the primary power conversion and control occur within the vehicle, the communication requirements between the EVSE and EV are less complex than in DC charging.

- **Low-Level PWM Communication**: AC charging typically employs low-level, Pulse Width Modulation (PWM)-based communication. This communication occurs over the Control Pilot (CP) pin of the charging connector.
- **Information Exchange**: The PWM signal's amplitude and duty ratio can be varied to convey different pieces of information. For instance, changes in these parameters are detected by the receiving end (EV or EVSE) and interpreted as specific data, such as start/shutdown indicators or maximum current allowances.
- **EV as Master**: In AC charging, the EV (via its BMS) is the master. It initiates and controls the charging process. The EVSE primarily responds to the EV's requests and provides information about its capabilities (e.g., maximum current it can supply).
- **Proximity Pilot (PP) for Initial Detection**: Before any PWM communication begins, the Proximity Pilot (PP) pin plays a crucial role in detecting the physical connection and verifying the charger plug.
- **Plug-in Recognition**: The EV first recognizes that a charger plug has been inserted via the PP pin.
- **Cable Capacity Detection**: The PP circuit allows the EV to determine if the connected charger plug is valid and, more importantly, what its maximum current carrying capacity is. This is achieved through a specific resistance value (R6) within the charger plug.

3. AC Type-2 and CCS2 Connector Standards

The physical interface for AC charging is standardized to ensure interoperability and safety.

- **AC Type-2 Connector**: This is a common standard for AC charging, particularly in Europe.
- **Pins**: An AC Type-2 charger plug and vehicle inlet connector typically include:
- L1, L2, L3, N: Live phases and Neutral for power transfer (larger pins).
- PE: Protective Earth for safety (larger pin).
- CP: Control Pilot for communication (smaller pin).

- PP: Proximity Pilot for plug detection and cable capacity (smaller pin).
- **Form Factor**: The charger plug and vehicle inlet connector have matching form factors to ensure a secure physical connection.
- **CCS2 (Combined Charging System Type 2)**: This is a versatile connector that integrates both AC Type-2 and DC charging capabilities into a single vehicle inlet.
- **Dual Functionality**: The top portion of a CCS2 vehicle inlet connector has the same form factor as an AC Type-2 inlet. This allows an AC Type-2 charger plug to connect to the top part of a CCS2 inlet for AC charging.
- **DC Charging**: For DC charging, the entire CCS2 connector is utilized, including additional DC+ and DC- pins, along with the CP, PP, and PE pins.
- **Convenience**: The CCS2 standard offers convenience by allowing a single vehicle inlet to support both slow AC charging (e.g., overnight at home) and fast DC charging (e.g., at public charging stations).

4. Proximity Pilot (PP) Circuit and Cable Detection

The PP circuit is a critical safety feature that allows the EV to identify the connected charger plug and its current carrying capabilities before power delivery begins.

- **Mechanism**: The charger plug contains a specific resistance (R6) connected between the PP and PE pins. The EV's internal circuit measures the voltage at the PP pin, which varies depending on the R6 value.
- **R6 Value and Current Rating**: Different R6 resistance values correspond to different maximum current ratings of the charging cable and plug. This allows the EV to know how much current it can safely draw.
- **Error Condition / Disconnected**: If the resistance between PP and PE is greater than 4500 Ω, or the voltage at the PP pin is greater than 4.37V, it indicates an error condition or a disconnected plug.
- **1500 Ω**: Corresponds to a 13A AC charging capacity (or CCS2 DC charging), with a conductor size of 1.5 mm^2. The rated voltage at the PP pin for this is 3.87V.
- **680 Ω**: Corresponds to a 20A AC charging capacity, with a conductor size of 2.5 mm^2. The rated voltage at the PP pin for this is 3.21V.
- **220 Ω**: Corresponds to a 32A AC charging capacity, with a conductor size of 6 mm^2. The rated voltage at the PP pin for this is 1.94V.
- **100 Ω**: Corresponds to a 63A AC (three-phase) charging capacity, with a conductor size of 16 mm^2. The rated voltage at the PP pin for this is 1.14V.

- **Error Condition**: If the resistance is less than 60 Ω, or the voltage at the PP pin is less than 0.76V, it indicates another error condition with the charger plug.
- **Tolerance Ranges**: In practice, these resistance values have tolerance ranges (e.g., 1500 Ω has a range of 1100-2460 Ω). The EV's controller uses these ranges to accurately identify the cable's capacity.
- **Safety Implication**: This mechanism ensures that the EV never attempts to draw more current than the connected cable and plug are safely rated for, preventing overheating and potential hazards.

5. Calculation Example for PP Voltage

To illustrate how the EV determines the R6 value and thus the cable capacity, let's consider the voltage divider circuit involving the PP pin.

The EV's internal circuit provides a +5V source. This voltage is applied across a series of resistors, including the R6 resistance from the charger plug.
Let's assume the following components within the EV's vehicle inlet circuit:
* R4 = 330 Ω
* R5 = 4700 Ω
* The R6 resistance from the charger plug.

When the charger plug is connected, the R6 resistance is placed in parallel with R5. The voltage at the PP pin (VPP) can then be calculated using a voltage divider formula.

Consider the case where R6 = 1500 Ω (for a 13A AC charger plug).
The equivalent resistance of R6 and R5 in parallel is:
$R_{parallel} = (R6 * R5) / (R6 + R5)$
$R_{parallel} = (1500 * 4700) / (1500 + 4700)$
$R_{parallel} = (7050000) / (6200)$
$R_{parallel} \approx 1137.09 \Omega$

Now, VPP is the voltage across $R_{parallel}$ in a voltage divider with R4:
$VPP = 5V * (R_{parallel} / (R4 + R_{parallel}))$
$VPP = 5V * (1137.09 / (330 + 1137.09))$
$VPP = 5V * (1137.09 / 1467.09)$
$VPP \approx 5V * 0.775$
$VPP \approx 3.87V$

This calculated voltage of approximately 3.87V matches the rated voltage for a 1500 Ω R6 value, as shown in the standards. The EV's controller measures this VPP and, by comparing it to predefined thresholds (often stored in a lookup table), identifies the maximum current capacity of the connected cable. This initial detection and

verification process is crucial before the more complex PWM communication over the CP pin can proceed to set up the actual charging session.

Key Takeaways

- [ ] AC charging involves power conversion primarily within the vehicle's onboard charger (AC-DC, isolated DC-DC).
- [ ] The ACVSE (charging station) provides AC power and manages communication, while the EV's BMS acts as the master controller.
- [ ] Communication between the EV and ACVSE for AC charging is low-level and PWM-based, occurring over the Control Pilot (CP) pin.
- [ ] The Proximity Pilot (PP) pin is responsible for detecting the charger plug and determining the maximum current capacity of the connected cable.
- [ ] The R6 resistance value within the charger plug's PP circuit signals the cable's current rating to the EV.
- [ ] Different R6 values correspond to specific maximum currents (e.g., 100Ω for 63A, 1500Ω for 13A) and conductor sizes.
- [ ] The CCS2 vehicle inlet connector can accommodate AC Type-2 plugs in its top portion for AC charging, offering versatility.
- [ ] Safety interlocks and the PP circuit are critical for preventing overcurrent and ensuring a safe charging connection.

Conclusion

AC Type-2 charging, while often perceived as the "slower" charging method, is a sophisticated system underpinned by precise communication and robust safety protocols. Its architecture, which places the primary power conversion unit (onboard charger) within the vehicle, simplifies the external EVSE and allows the vehicle's Battery Management System to maintain ultimate control over the charging process. This master-slave relationship, coupled with low-level PWM communication over the Control Pilot pin, ensures that charging parameters are optimized for battery health and longevity.

Crucially, the Proximity Pilot (PP) circuit, with its ingenious use of the R6 resistance, provides an essential layer of safety. By allowing the EV to automatically detect the connected cable's current carrying capacity, it prevents dangerous overloads and ensures that only appropriate power levels are requested. The versatility of standards like CCS2, which integrates AC Type-2 compatibility, further streamlines the user experience by offering a single vehicle inlet for diverse charging needs. As EV adoption continues to grow, understanding these foundational AC charging

principles becomes increasingly vital for developing and maintaining a reliable and safe charging ecosystem.